USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 5 LAB - First Flowgraphs
Lab Packet
Spring 2027

Name: _____ **Section:** _____ **Date:** _____

Learning Objective 2.3: I can install GNU Radio and use GNU Radio Companion (GRC) to build, inspect, and run a basic flowgraph.
Learning Objective 2.4: I can capture a real RF signal using an SDR and interpret its time- and frequency-domain representations.

This packet is the L5 Lab, in three parts, and you turn it in once, complete, before Lesson 6. Do **Part 1** on your own before the lab period: install GNU Radio and make the predictions you will check at the bench. **Part 2** is your own measurements, with your own laptop and RTL-SDR, recorded in class. Do **Part 3** on your own after the lab: compare the measurements with the predictions and answer the analysis questions.

Grading. One submission, graded against Objectives 2.3 and 2.4 on the course's 1 / 0 scale, with revision allowed under course policy. Paste screenshots where a question asks for one.

Receive only. Nothing in this lab transmits. The RTL-SDR cannot transmit, and the tone flowgraph never leaves your laptop.

The flowgraphs and the Python script are on the L5 lesson page: `L05-tone.grc`, `L05-fm-spectrum.grc`, and `L05-psd.py`.

Part 1: Pre-Lab (individual, before the lab period)

1. **Install and verify.** Install radioconda as the L5 lesson page describes, plug in your RTL-SDR, and run the commands below from the radioconda prompt.

   (a) Run `gnuradio-config-info --version` and record the version.

   version = _____

   (b) Run `rtl_test` and record the tuner it reports. Stop it with Ctrl-C after a few seconds.

   tuner = _____

   (c) Open `L05-tone.grc` in GRC, run it, and paste a screenshot of the running flowgraph's window.

2. Predictions. You will check each of these at the bench.

(a) The tone flowgraph samples at $f_s = 32$ kHz. Predict where the frequency sink shows each tone.

Tone (kHz)	1	12	20	30	-25
Predicted (kHz)					

(b) The FM flowgraph samples at 2.4 MS/s centered on 100.1 MHz. Find the band the display shows and the width of one bin of its 2048-point FFT.

band =

bin =

(c) Predict the size of a 2 s recording at 2.4 MS/s.

Part 2: Measurements (your own laptop and RTL-SDR, in class)

3. **Build the tone flowgraph.** Start GRC with an empty canvas. Do not open `L05-tone.grc`: build it yourself from the Options block, a `samp_rate` variable of 32000, a QT GUI Range named `freq` from -48000 to 48000, a complex Signal Source, a Throttle, a QT GUI Time Sink, and a QT GUI Frequency Sink.

   (a) Paste a screenshot of your flowgraph on the GRC canvas. Every connection must be black, not red.

   (b) Set each tone with the slider and record where the spike appears.

Tone (kHz)	1	12	20	30	-25
Measured (kHz)					

4. The FM band. Connect the antenna to the RTL-SDR and run `L05-fm-spectrum.grc`.

(a) Tune across 88 to 108 MHz and record the three strongest stations: the frequency, the peak level on the display, and the approximate width.

	Frequency (MHz)	Peak (dB)	Width (kHz)
Station 1			
Station 2			
Station 3			

(b) Find the DC spike. Tune so that a station sits on it, then tune 300 kHz away. Describe what moves and what does not.

(c) With the strongest station on screen, set the gain to each value and record the noise floor away from any station. Mark the first setting where extra signals appear between stations.

Gain (dB)	0	20	30	40	49
Floor (dB)					
Spurs?					

5. Record and plot. Center the strongest station slightly off the DC spike and set the gain you chose in Question 4(c).

(a) Enable the File Sink, run the flowgraph for about 2 s, stop it, and record the file size.

size =

(b) Run the plotting script with your center frequency, and paste the plot:

```
python L05-psd.py fm_capture.cfile --fs 2.4e6 --fc <center Hz>
```

Part 3: Writeup (individual, after the lab)

6. Analysis.

(a) Compare your measured aliases in Question 3(b) with the predictions in Question 2(a). Explain why a 20 kHz tone and a -12 kHz tone produce the same samples at 32 kHz.

(b) Your strongest station's peak sits some number of dB above the noise floor on the display. Is that the SNR an FM demodulator would see? Explain, and estimate the correction for a 200 kHz station with 1.17 kHz bins.

(c) State the gain you would use to receive weak stations in this room, and justify it from your table in Question 4(c).

Documentation: _____